



WHERE LEGAL EXPERTISE MEETS AI TECHNOLOGY

Autonomous Evidence Discovery & Regulatory Mapping for Digital Trade

Find the law, prove the clause, automatically.

Round 1 economies:

Singapore

Australia

Malaysia

Veritrade Team - Foreign Trade University, Vietnam

Contact: minhtc@ftu.edu.vn

**Assoc. Prof. Dr.
Dao Ngoc Tien**
Policy Lead

**Assoc. Prof. Dr.
Nguyen Ngoc Ha**
Legal Research Lead

Dr. Nguyen Thu Hang
Policy and Legal Advisor

MSc. Tran Cong Minh
Technical Lead

Vo Minh Ngoc
Legal - Tech Researcher
Developer

FTU is Vietnam's leading university of international trade: this team pairs trade-policy and legal training with engineering execution. VeriTrade is built by the exact profile of the people who will use it, policy researchers who can also read the code.

Executive Summary

Given only an **economy** and a **pillar**, VeriTrade autonomously finds the laws on the official portals, reads them (including scanned PDFs), and returns every provision mapped to its RDTII indicator: verbatim quote, citation, source URL, confidence score. **No seed URLs. No hardcoded law names. Every row auditable.**

CER 1.11%

measured OCR error on scanned gazettes
(official bar: 5%)

**~\$0.07 /
economy**

LLM cost per run (\$0.00 open-weight stack,
CPU-only)

**8–18 min /
economy**

measured live-run wall-clock, both pillars,
default stack

measured; recorded in every output JSON.
Repeat runs near-instant from cache

What ESCAP's own orientation describes as "hours of manual database searching" per indicator becomes a reproducible 8–18-minute run per economy, making regular refreshes across the full RDTII coverage feasible, not just the 3 Round-1 economies.

For every economy, a researcher must **find** the right statutes across bespoke government portals, **read** each operative clause (often a scanned PDF, sometimes in another language), **interpret** it against the indicator, and **cite** it precisely. ESCAP's own orientation puts the search step alone at "hours of manual database searching", and that is one indicator; an economy needs 9 indicators × several statutes × three file formats. This hackathon alone spans 11 economies (3 in Round 1, 8 more in the Finals), each needing periodic refreshes. **The manual process cannot scale, and a second reviewer cannot reproduce it.**

And naive automation fails: the portals fight back

Every portal is different, none is friendly

SG's search is token/JS-gated; MY is barely indexed by Google; AU is an Angular app whose law text isn't in the HTML a crawler sees.

Bot-protected at the handshake

WAFs fingerprint the TLS handshake itself, a plain Python crawler gets HTTP 403 before sending a single header.

Australia's full-text search is broken

The official API's \$search returns errors, title-only search is a forced constraint; topical relevance needs a separate content lane.

Mixed formats & languages

HTML, text PDFs, scanned gazettes; Malay authoritative with English reference; finals add Thai, Chinese, Russian, Lao.

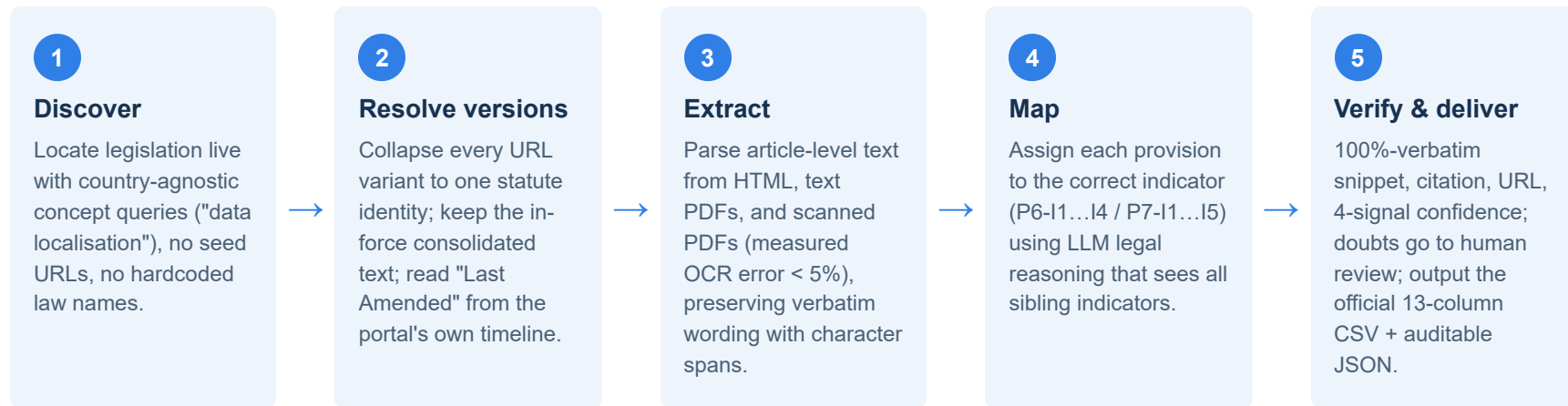
The evidence is scattered

P7-I5 (government access) lives in criminal-procedure, telecom and tax law, a title search never finds the Companies Act's record-storage clause.

Verification burden

A citation is only trustworthy if the quoted text truly appears in the official source, the core anti-hallucination requirement.

Automate the professional legal researcher's workflow, end-to-end, reproducibly, on a standard CPU, with no vendor lock-in.



DESIGN PRINCIPLES

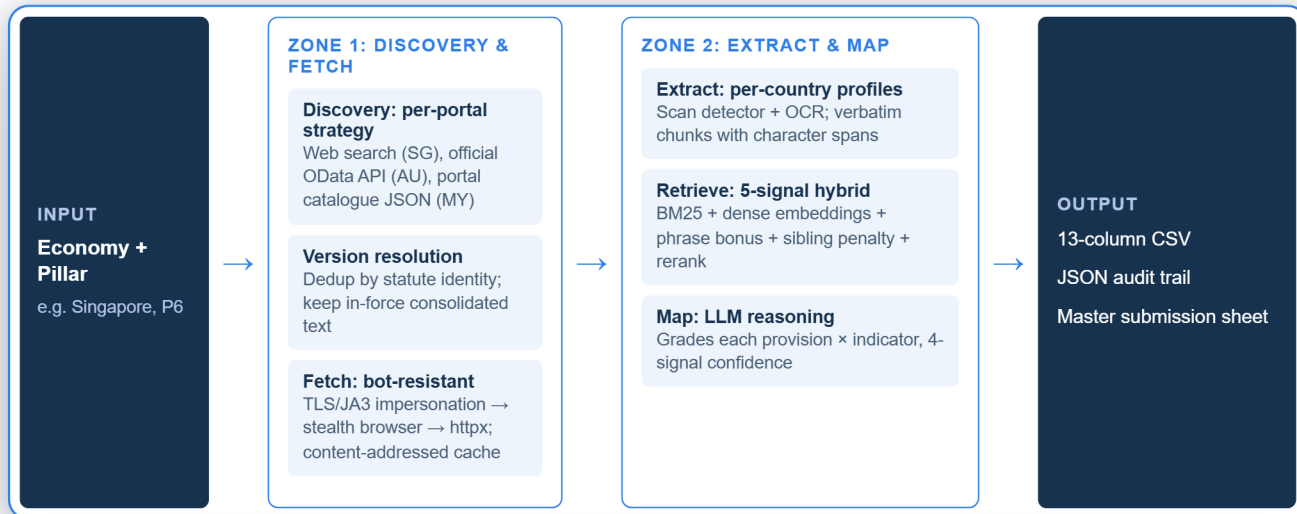
Auditable over opaque

Reproducible over one-off

Swappable over lock-in

Human-review safety net

System Architecture



CONFIDENCE ROUTING

≥ 0.85, auto-accept

0.60 – 0.85, human review

< 0.60, quarantine

IN PLAIN TERMS Type a country and a topic. The system finds the laws, reads them, quotes the exact sentence that proves each answer, and hands anything uncertain to a human. Live and sample runs share the same code, so a sample run honestly previews a live one.

VERITRADE PIPELINE

Economy + Pillar

e.g. Singapore · P6

ZONE 1 · DISCOVERY & FETCH

Discover

per-portal, no seed URLs

Resolve versions

one statute identity, in-force

Fetch

TLS/JA3 → stealth browser

ZONE 2 · EXTRACT & MAP

Extract

scan detect · OCR · verbatim

Retrieve

5-signal hybrid shortlist

Map

sibling-aware LLM judge

Confidence

accept / review / quarantine

ZONE 3 · SCORING

Score

0 · 0.5 · 1 raw score

13-column CSV

+ JSON audit trail

one run, input to checkable evidence

Evaluation & Performance



OCR quality (CER)

PASS

1.11%

measured error vs the 5% pass bar

Citation fidelity

100%

verbatim by construction, every snippet must occur in the fetched source

Coverage

9/9

every indicator gets a row or an explicit "no evidence" placeholder

Discovery precision

14 laws, 0 junk

Australia P6: all in-force and on-topic; the AU SPA shell is detected and suppressed: 0 bogus provisions

Cost per economy

~\$0.07 or \$0.00

measured LLM cost, both pillars (~360 calls), \$0.00 on an open-weight stack (Ollama + Tesseract)

Response time

10–17 min

per economy, live, CPU-only (extract 44%, embed 37%, LLM 14%), embedding disk-cache makes repeats 16× faster

Transparency note: the LLM grader is stochastic, borderline clauses used to flip between runs. Three mitigations: a cross-model majority panel on borderline rejections, a review queue as the human safety net, and a result cache that freezes a chosen run for judging. Output follows the judges' Q&A exactly: never-amended laws read 'Original' in Last Amended, and no-evidence rows carry N/A confidence, never silent blanks.

Multilingual: multilingual embeddings + grade-all protect non-English recall; Malaysia's bilingual corpus handled.

VERITRADE PIPELINE

Economy + Pillar

e.g. Singapore · P6

ZONE 1 · DISCOVERY & FETCH

Discover

per-portal, no seed URLs

Resolve versions

one statute identity, in-force

Fetch

TLS/JA3 → stealth browser

ZONE 2 · EXTRACT & MAP

Extract

scan detect · OCR · verbatim

Retrieve

5-signal hybrid shortlist

Map

sibling-aware LLM judge

Confidence

accept / review / quarantine

ZONE 3 · SCORING

Score

0 · 0.5 · 1 raw score

13-column CSV

quote · citation · URL · confidence

every row is verifiable by a second reviewer

Backend Logic (1/2): Discover, Extract, Retrieve



1. Discover

- Two query lanes: name fragments ("privacy act") + obligation phrases ("data must be stored in"), the latter only where full-text search works
- Version resolution: URL variants collapse to one statute identity; "Last Amended" read from the portal's own timeline, never guessed
- Repealed / bill hits dropped; AU re-verified in-force by register title-id

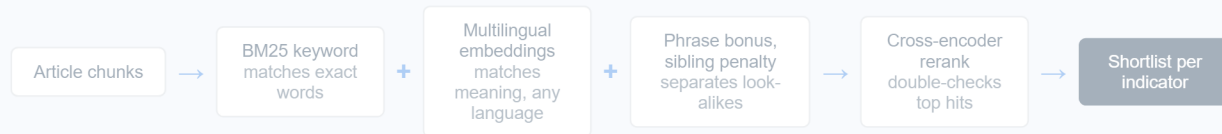
2. Fetch

- Browser-handshake impersonation (TLS/JA3) → stealth-browser escalation → simple-fetch fallback
- Polite per-host delays; content-addressed cache (same bytes never fetched twice)
- Landing pages with embedded viewers resolved to the actual PDF

3. Extract

- Text-density detector catches "secretly scanned" PDFs → raster OCR, CER measured
- SG/MY: split at margin-numbered sections ("11.—(1)"); AU: headings found by their bold typeface, their drafting has no "Section" keyword
- Statistical page-chrome stripping → verbatim chunks with character spans

FINDING THE RELEVANT SECTIONS, 5-SIGNAL HYBRID RETRIEVAL



Corpus ≤ 80 provisions? **Every provision is graded**, retrieval is a signal, not a gate. Larger corpora: shortlist of 20–40 per indicator with **per-law reserved slots**. 1,200 provisions ⇒ ~360 LLM calls, not 10,800.

IN PLAIN TERMS Like a good librarian, the system shortlists passages by exact words and by meaning, in any language, then a lawyer-like reader makes the final call on each one.

VERITRADE PIPELINE

Economy + Pillar

e.g. Singapore · P6

ZONE 1 · DISCOVERY & FETCH

Discover

two lanes: name + obligation phrase

Resolve versions

one statute identity, in-force

Fetch

TLS/JA3 → stealth browser

ZONE 2 · EXTRACT & MAP

Extract

scan detect · OCR · verbatim

Retrieve

5-signal hybrid shortlist

Map

sibling-aware LLM judge

Confidence

accept / review / quarantine

ZONE 3 · SCORING

Score

0 · 0.5 · 1 raw score

13-column CSV

+ JSON audit trail

"data must be stored in" finds the Companies Act

Backend Logic (1/2): Discover, Extract, Retrieve



1. Discover

- Two query lanes: name fragments ("privacy act") + obligation phrases ("data must be stored in"), the latter only where full-text search works
- Version resolution: URL variants collapse to one statute identity; "Last Amended" read from the portal's own timeline, never guessed
- Repealed / bill hits dropped; AU re-verified in-force by register title-id

2. Fetch

- Browser-handshake impersonation (TLS/JA3) → stealth-browser escalation → simple-fetch fallback
- Polite per-host delays; content-addressed cache (same bytes never fetched twice)
- Landing pages with embedded viewers resolved to the actual PDF

3. Extract

- Text-density detector catches "secretly scanned" PDFs → raster OCR, CER measured
- SG/MY: split at margin-numbered sections ("11.—(1)"); AU: headings found by their bold typeface, their drafting has no "Section" keyword
- Statistical page-chrome stripping → verbatim chunks with character spans

FINDING THE RELEVANT SECTIONS, 5-SIGNAL HYBRID RETRIEVAL



IN PLAIN TERMS Like a good librarian, the system shortlists passages by exact words and by meaning, in any language, then a lawyer-like reader makes the final call on each one.

VERITRADE PIPELINE

Economy + Pillar

e.g. Singapore · P6

ZONE 1 · DISCOVERY & FETCH

Discover

per-portal, no seed URLs

Resolve versions

one statute identity, in-force only

Fetch

TLS/JA3 → stealth browser

ZONE 2 · EXTRACT & MAP

Extract

scan detect · OCR · verbatim

Retrieve

5-signal hybrid shortlist

Map

sibling-aware LLM judge

Confidence

accept / review / quarantine

ZONE 3 · SCORING

Score

0 · 0.5 · 1 raw score

13-column CSV

+ JSON audit trail

"Last Amended" read from the portal's own timeline, never guessed

Backend Logic (1/2): Discover, Extract, Retrieve



1. Discover

- Two query lanes: name fragments ("privacy act") + obligation phrases ("data must be stored in"), the latter only where full-text search works
- Version resolution: URL variants collapse to one statute identity; "Last Amended" read from the portal's own timeline, never guessed
- Repealed / bill hits dropped; AU re-verified in-force by register title-id

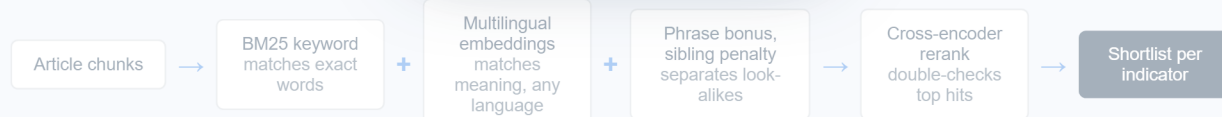
2. Fetch

- Browser-handshake impersonation (TLS/JA3) → stealth-browser escalation → simple-fetch fallback
- Polite per-host delays; content-addressed cache (same bytes never fetched twice)
- Landing pages with embedded viewers resolved to the actual PDF

3. Extract

- Text-density detector catches "secretly scanned" PDFs → raster OCR, CER measured
- SG/MY: split at margin-numbered sections ("11.—(1)"); AU: headings found by their bold typeface, their drafting has no "Section" keyword
- Statistical page-chrome stripping → verbatim chunks with character spans

FINDING THE RELEVANT SECTIONS, 5-SIGNAL HYBRID RETRIEVAL



Corpus ≤ 80 provisions? **Every provision is graded**, retrieval is a signal, not a gate. Larger corpora: shortlist of 20–40 per indicator with **per-law reserved slots**. 1,200 provisions ⇒ ~360 LLM calls, not 10,800.

IN PLAIN TERMS Like a good librarian, the system shortlists passages by exact words and by meaning, in any language, then a lawyer-like reader makes the final call on each one.

VERITRADE PIPELINE

Economy + Pillar

e.g. Singapore · P6

ZONE 1 · DISCOVERY & FETCH

Discover

per-portal, no seed URLs

Resolve versions

one statute identity, in-force

Fetch

handshake impersonation, then stealth browser

ZONE 2 · EXTRACT & MAP

Extract

scan detect · OCR · verbatim

Retrieve

5-signal hybrid shortlist

Map

sibling-aware LLM judge

Confidence

accept / review / quarantine

ZONE 3 · SCORING

Score

0 · 0.5 · 1 raw score

13-column CSV

+ JSON audit trail

WAFs block the knock, not the words

Competitive Advantage



No baked corpus

The obligation-phrase lane finds laws by what their clauses say; the MY adapter reads the portal's own catalogue. New economy = one portal domain + one entry strategy.

Verifiable by design

Verbatim snippets with character spans, grounding checks, source URLs, 4-signal confidence, full JSON audit trail, a second reviewer can check every row.

Legal reliability, encoded three times

Sibling-aware prompting encodes the RDTII distinguish-from rules; the sibling penalty applies them again at retrieval; conservative defaults + a review queue prevent confident-but-wrong mappings, and a cross-model panel of three independent LLMs re-judges borderline rejections by majority vote.

Survives real portals

Broken official search (AU), a portal Google can't see (MY), robot-blocking firewalls, pages that load their text with JavaScript, "secretly scanned" PDFs, bilingual catalogues, each has a specific, tested countermeasure. Most pipelines demo on clean HTML; ours demos on the real thing.

Zone 3 scoring, polarity included

Official 0/0.5/1 raw scores, including the counter-intuitive rule that a strong privacy framework must score as less restrictive (the inverted polarity of measures 7.1/7.2). A detail most teams miss.

No lock-in, runs anywhere

LLM and OCR swap via one config value (OpenRouter, Anthropic, OpenAI, Gemini, local Ollama). CPU-only, ~\$0.07 per economy, \$0.00 open-weight; caches make repeats free.

VERITRADE PIPELINE

Economy + Pillar

e.g. Singapore · P6

ZONE 1 · DISCOVERY & FETCH

Discover

per-portal, no seed URLs

Resolve versions

one statute identity, in-force

Fetch

broken search · invisible portal · JS shell

ZONE 2 · EXTRACT & MAP

Extract

scan detect · OCR · verbatim

Retrieve

5-signal hybrid shortlist

Map

sibling-aware LLM judge

Confidence

accept / review / quarantine

ZONE 3 · SCORING

Score

0 · 0.5 · 1 raw score

13-column CSV

+ JSON audit trail

the silent failure is a crawler that "succeeds" on menus: 0 here

Backend Logic (1/2): Discover, Extract, Retrieve



1. Discover

- Two query lanes: name fragments ("privacy act") + obligation phrases ("data must be stored in"), the latter only where full-text search works
- Version resolution: URL variants collapse to one statute identity; "Last Amended" read from the portal's own timeline, never guessed
- Repealed / bill hits dropped; AU re-verified in-force by register title-id

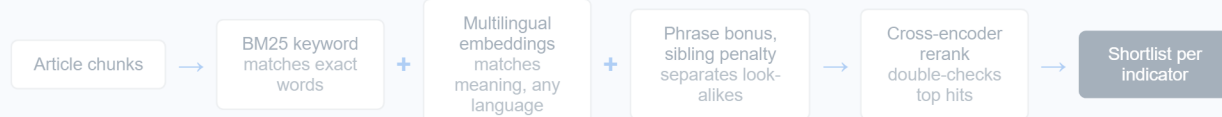
2. Fetch

- Browser-handshake impersonation (TLS/JA3) → stealth-browser escalation → simple-fetch fallback
- Polite per-host delays; content-addressed cache (same bytes never fetched twice)
- Landing pages with embedded viewers resolved to the actual PDF

3. Extract

- Text-density detector catches "secretly scanned" PDFs → raster OCR, CER measured
- SG/MY: split at margin-numbered sections ("11.—(1)"); AU: headings found by their bold typeface, their drafting has no "Section" keyword
- Statistical page-chrome stripping → verbatim chunks with character spans

FINDING THE RELEVANT SECTIONS, 5-SIGNAL HYBRID RETRIEVAL



Corpus ≤ 80 provisions? **Every provision is graded**, retrieval is a signal, not a gate. Larger corpora: shortlist of 20–40 per indicator with **per-law reserved slots**. 1,200 provisions ⇒ ~360 LLM calls, not 10,800.

IN PLAIN TERMS Like a good librarian, the system shortlists passages by exact words and by meaning, in any language, then a lawyer-like reader makes the final call on each one.

VERITRADE PIPELINE

Economy + Pillar

e.g. Singapore · P6

ZONE 1 · DISCOVERY & FETCH

Discover

per-portal, no seed URLs

Resolve versions

one statute identity, in-force

Fetch

TLS/JA3 → stealth browser

ZONE 2 · EXTRACT & MAP

Extract

scan detector, then OCR: CER 1.11%

Retrieve

5-signal hybrid shortlist

Map

sibling-aware LLM judge

Confidence

accept / review / quarantine

ZONE 3 · SCORING

Score

0 · 0.5 · 1 raw score

13-column CSV

+ JSON audit trail

measured against ground truth, not claimed

Backend Logic (1/2): Discover, Extract, Retrieve



1. Discover

- Two query lanes: name fragments ("privacy act") + obligation phrases ("data must be stored in"), the latter only where full-text search works
- Version resolution: URL variants collapse to one statute identity; "Last Amended" read from the portal's own timeline, never guessed
- Repealed / bill hits dropped; AU re-verified in-force by register title-id

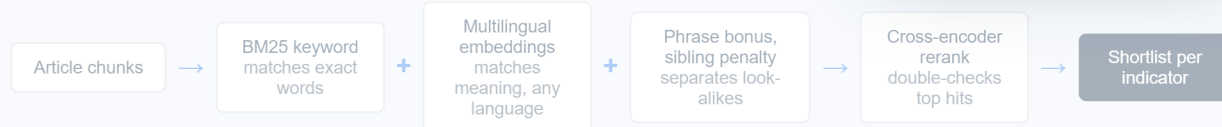
2. Fetch

- Browser-handshake impersonation (TLS/JA3) → stealth-browser escalation → simple-fetch fallback
- Polite per-host delays; content-addressed cache (same bytes never fetched twice)
- Landing pages with embedded viewers resolved to the actual PDF

3. Extract

- Text-density detector catches "secretly scanned" PDFs → raster OCR, CER measured
- SG/MY: split at margin-numbered sections ("11.—(1)"); AU: headings found by their bold typeface, their drafting has no "Section" keyword**
- Statistical page-chrome stripping → verbatim chunks with character spans

FINDING THE RELEVANT SECTIONS, 5-SIGNAL HYBRID RETRIEVAL



Corpus ≤ 80 provisions? **Every provision is graded**, retrieval is a signal, not a gate. Larger corpora: shortlist of 20–40 per indicator with **per-law reserved slots**. 1,200 provisions ⇒ ~360 LLM calls, not 10,800.

IN PLAIN TERMS Like a good librarian, the system shortlists passages by exact words and by meaning, in any language, then a lawyer-like reader makes the final call on each one.

VERITRADE PIPELINE

Economy + Pillar

e.g. Singapore · P6

ZONE 1 · DISCOVERY & FETCH

Discover

per-portal, no seed URLs

Resolve versions

one statute identity, in-force

Fetch

TLS/JA3 → stealth browser

ZONE 2 · EXTRACT & MAP

Extract

one splitting profile per drafting style

Retrieve

5-signal hybrid shortlist

Map

sibling-aware LLM judge

Confidence

accept / review / quarantine

ZONE 3 · SCORING

Score

0 · 0.5 · 1 raw score

13-column CSV

+ JSON audit trail

verbatim is machine-checkable, not a promise

Backend Logic (1/2): Discover, Extract, Retrieve



1. Discover

- Two query lanes: name fragments ("privacy act") + obligation phrases ("data must be stored in"), the latter only where full-text search works
- Version resolution: URL variants collapse to one statute identity; "Last Amended" read from the portal's own timeline, never guessed
- Repealed / bill hits dropped; AU re-verified in-force by register title-id

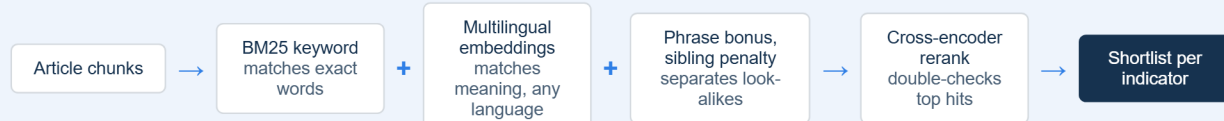
2. Fetch

- Browser-handshake impersonation (TLS/JA3) → stealth-browser escalation → simple-fetch fallback
- Polite per-host delays; content-addressed cache (same bytes never fetched twice)
- Landing pages with embedded viewers resolved to the actual PDF

3. Extract

- Text-density detector catches "secretly scanned" PDFs → raster OCR, CER measured
- SG/MY: split at margin-numbered sections ("11.—(1)"); AU: headings found by their bold typeface, their drafting has no "Section" keyword
- Statistical page-chrome stripping → verbatim chunks with character spans

FINDING THE RELEVANT SECTIONS, 5-SIGNAL HYBRID RETRIEVAL



Corpus ≤ 80 provisions? **Every provision is graded**, retrieval is a signal, not a gate. Larger corpora: shortlist of 20–40 per indicator with **per-law reserved slots**. 1,200 provisions ⇒ ~360 LLM calls, not 10,800.

IN PLAIN TERMS Like a good librarian, the system shortlists passages by exact words and by meaning, in any language, then a lawyer-like reader makes the final call on each one.

VERITRADE PIPELINE

Economy + Pillar

e.g. Singapore · P6

ZONE 1 · DISCOVERY & FETCH

Discover

per-portal, no seed URLs

Resolve versions

one statute identity, in-force

Fetch

TLS/JA3 → stealth browser

ZONE 2 · EXTRACT & MAP

Extract

scan detect · OCR · verbatim

Retrieve

BM25 + dense + phrase + sibling + rerank

Map

sibling-aware LLM judge

Confidence

accept / review / quarantine

ZONE 3 · SCORING

Score

0 · 0.5 · 1 raw score

13-column CSV

+ JSON audit trail

retrieval is a signal, never a gate

Backend Logic (2/2): Map & Verify



SIBLING-AWARE LLM MAPPING

For each (provision, indicator) pair the model receives the indicator's **legal test**, its scope, and **every sibling indicator's test**, then reasons in explicit steps: operative rule → does it satisfy the target? → is a sibling a better fit?

Ambiguity gets a conservative default (*unsure means reject*) and borderline rows go to the review queue, never silently accepted.

CROSS-MODEL SECOND OPINION

Borderline rejections don't rest on one model: a second, independent model re-grades the same provision in a context-free call, and a third breaks ties, majority (2-1) decides (deepseek-v4-flash primary; gemini-2.5-flash + gpt-4o-mini panel). Every vote is logged in the audit trail.

IN PLAIN TERMS The AI reads each clause like a lawyer holding the full rulebook of definitions, and when unsure, it asks a human instead of guessing. It can never invent a quote: every citation is copied from the official law and machine-checked to exist there.

WHY CITATIONS CANNOT BE HALLUCINATED

- 1 The snippet is copied from the fetched source, the LLM never writes it
- 2 A grounding check confirms it occurs in the source document
- 3 A topical guard flags snippets with none of the pillar's vocabulary
- 4 The source URL + surrounding context enable one-click human verification
- 5 Every row carries a confidence score; low scores go to review or quarantine

EXAMPLE: INPUT: SINGAPORE, PILLAR 6

Law	Personal Data Protection Act 2012 (Act 26 of 2012)
Indicator	P6-I4 , conditional cross-border flow
Section	Section 26
Verbatim snippet	"An organisation shall not transfer any personal data to a country or territory outside Singapore except in accordance with requirements prescribed under this Act..."
Rationale	Transfer is conditionally allowed, not banned, so P6-I4, not P6-I1.
Source	sso.agc.gov.sg/Act/PDPA2012
Confidence	0.82 → routed to human review

The sibling-aware prompt makes exactly this distinction: **conditional flow (I4)**, not a **ban (I1)**, the most common human-mapping confusion.

VERITRADE PIPELINE

Economy + Pillar

e.g. Singapore · P6

ZONE 1 · DISCOVERY & FETCH

Discover

per-portal, no seed URLs

Resolve versions

one statute identity, in-force

Fetch

TLS/JA3 → stealth browser

ZONE 2 · EXTRACT & MAP

Extract

scan detect · OCR · verbatim

Retrieve

5-signal hybrid shortlist

Map

the legal test plus every sibling test

Confidence

accept / review / quarantine

ZONE 3 · SCORING

Score

0 · 0.5 · 1 raw score

13-column CSV

+ JSON audit trail

borderline rejections go to a cross-model 2-1 panel

Competitive Advantage



No baked corpus

The obligation-phrase lane finds laws by what their clauses say; the MY adapter reads the portal's own catalogue. New economy = one portal domain + one entry strategy.

Verifiable by design

Verbatim snippets with character spans, grounding checks, source URLs, 4-signal confidence, full JSON audit trail, a second reviewer can check every row.

Legal reliability, encoded three times

Sibling-aware prompting encodes the RDTII distinguish-from rules; the sibling penalty applies them again at retrieval; conservative defaults + a review queue prevent confident-but-wrong mappings, and a cross-model panel of three independent LLMs re-judges borderline rejections by majority vote.

Survives real portals

Broken official search (AU), a portal Google can't see (MY), robot-blocking firewalls, pages that load their text with JavaScript, "secretly scanned" PDFs, bilingual catalogues, each has a specific, tested countermeasure. Most pipelines demo on clean HTML; ours demos on the real thing.

Zone 3 scoring, polarity included

Official 0/0.5/1 raw scores, including the counter-intuitive rule that a strong privacy framework must score as less restrictive (the inverted polarity of measures 7.1/7.2). A detail most teams miss.

No lock-in, runs anywhere

LLM and OCR swap via one config value (OpenRouter, Anthropic, OpenAI, Gemini, local Ollama). CPU-only, ~\$0.07 per economy, \$0.00 open-weight; caches make repeats free.

VERITRADE PIPELINE

Economy + Pillar

e.g. Singapore · P6

ZONE 1 · DISCOVERY & FETCH

Discover

per-portal, no seed URLs

Resolve versions

one statute identity, in-force

Fetch

TLS/JA3 → stealth browser

ZONE 2 · EXTRACT & MAP

Extract

scan detect · OCR · verbatim

Retrieve

5-signal hybrid shortlist

Map

sibling-aware LLM judge

Confidence

accept / review / quarantine

ZONE 3 · SCORING

Score

including the inverted polarity of 7.1 / 7.2

13-column CSV

+ JSON audit trail

the optional zone, implemented

Evaluation & Performance



OCR quality (CER)

PASS

1.11%

measured error vs the 5% pass bar

Citation fidelity

100%

verbatim by construction, every snippet must occur in the fetched source

Coverage

9/9

every indicator gets a row or an explicit "no evidence" placeholder

Discovery precision

14 laws, 0 junk

Australia P6: all in-force and on-topic; the AU SPA shell is detected and suppressed: 0 bogus provisions

Cost per economy

~\$0.07 or \$0.00

measured LLM cost, both pillars (~360 calls), \$0.00 on an open-weight stack (Ollama + Tesseract)

Response time

10–17 min

per economy, live, CPU-only (extract 44%, embed 37%, LLM 14%), embedding disk-cache makes repeats 16× faster

Transparency note: the LLM grader is stochastic, borderline clauses used to flip between runs. Three mitigations: a cross-model majority panel on borderline rejections, a review queue as the human safety net, and a result cache that freezes a chosen run for judging. Output follows the judges' Q&A exactly: never-amended laws read "Original" in Last Amended, and no-evidence rows carry N/A confidence, never silent blanks.

Multilingual: multilingual embeddings + grade-all protect non-English recall; Malaysia's bilingual corpus handled.

VERITRADE PIPELINE

Economy + Pillar

e.g. Singapore · P6

ZONE 1 · DISCOVERY & FETCH

Discover

per-portal, no seed URLs

Resolve versions

one statute identity, in-force

Fetch

TLS/JA3 → stealth browser

ZONE 2 · EXTRACT & MAP

Extract

scan detect · OCR · verbatim

Retrieve

5-signal hybrid shortlist

Map

sibling-aware LLM judge

Confidence

accept / review / quarantine

ZONE 3 · SCORING

Score

0 · 0.5 · 1 raw score

13-column CSV

+ JSON audit trail

measured, recorded in every output file

Scalability & Impact

SCALING OUT

More economies

Per-economy work is pluggable by design: one entry strategy + one extraction profile for the drafting convention ("Điều 5", "第五条", "มาตรา ๕"), the adapter pattern already proven three times.

More languages

Embeddings already cover Thai / Chinese / Russian; a multilingual cross-encoder (bge-reranker-v2-m3) is one config line; dual-engine translation cross-check with bilingual snippets planned.

More domains

Indicators are data-driven, add other RDTII pillars by defining their legal tests, with zero pipeline changes.

Real-time monitoring

Scheduled re-crawls diff against cached results to flag amended or new laws automatically.

ROUND 2 ROADMAP

Weeks 1–2

Discovery entry strategies + extraction profiles for the Finals economies (Thailand, China, India, Indonesia, Russia, Lao PDR, Mongolia, Timor-Leste): the same adapter pattern already proven 3× on SG/AU/MY.

Weeks 2–3

Multilingual reranker swap (bge-reranker-v2-m3, one config line) · per-page scan detection for mixed text/scan gazettes · dual-engine translation cross-check.

Week 4

Full 8-economy validation against the answer key · reviewer dashboard hardening · measured cost/time report per economy.

VALUE CREATED

ESCAP

Faster, reproducible, auditable RDTII refreshes at a fraction of the manual cost.

Governments & businesses

Quicker access to authoritative, cited digital-trade rules; lower compliance-research burden.

Researchers

A transparent, swappable, open-source base to extend.

SDG ALIGNMENT

SDG 8 Lowers the cost of digital-trade integration analysis, supporting evidence-based policy that reduces trade friction.

SDG 16 Every mapping is a verbatim, source-linked, auditable citation: transparent access to law and accountable regulatory evidence (targets 16.6, 16.10).

SDG 17 Builds regional data and monitoring capacity (17.18); Apache-2.0 so member states can adopt and extend it.



VERITRADE PIPELINE

Economy + Pillar

e.g. Singapore · P6

ZONE 1 · DISCOVERY & FETCH

+ Discover

+1 entry strategy per economy

Resolve versions

one statute identity, in-force

Fetch

TLS/JA3 → stealth browser

ZONE 2 · EXTRACT & MAP

+ Extract

+1 extraction profile per drafting style

Retrieve

5-signal hybrid shortlist

Map

sibling-aware LLM judge

Confidence

accept / review / quarantine

ZONE 3 · SCORING

Score

0 · 0.5 · 1 raw score

13-column CSV

+ JSON audit trail

every item plugs into a slot the system already has

References



OFFICIAL LEGAL SOURCES

Singapore Statutes Online	sso.agc.gov.sg
Federal Register of Legislation (AU)	legislation.gov.au
Laws of Malaysia (AGC)	lom.agc.gov.my
UN ESCAP RDTII 2.1 Methodology	provided by organisers

OPEN-SOURCE LIBRARIES

Scrapling httpx BeautifulSoup MarkItDown
RapidOCR PaddleOCR sentence-transformers
rank_bm25 pdfplumber Streamlit pydantic

All permissively licensed: MIT / Apache-2.0 / BSD.

AI MODELS

LLM	DeepSeek V4 Flash via OpenRouter, swappable
Embeddings	multilingual MiniLM-L12-v2
Reranker	ms-marco MiniLM-L-6-v2

SEARCH & REPOSITORY

Search API	Serper.dev, DuckDuckGo / Mojeek fallback
Repository	github.com/ftulabs/law-v2.0
License	Apache-2.0

VeriTrade: find the law, prove the clause, automatically.

VERITRADE PIPELINE

Economy + Pillar

e.g. Singapore · P6

ZONE 1 · DISCOVERY & FETCH

Discover

per-portal, no seed URLs

Resolve versions

one statute identity, in-force

Fetch

TLS/JA3 → stealth browser

ZONE 2 · EXTRACT & MAP

Extract

scan detect · OCR · verbatim

Retrieve

5-signal hybrid shortlist

Map

sibling-aware LLM judge

Confidence

accept / review / quarantine

ZONE 3 · SCORING

Score

0 · 0.5 · 1 raw score

13-column CSV

+ JSON audit trail

fully open stack, swappable by config